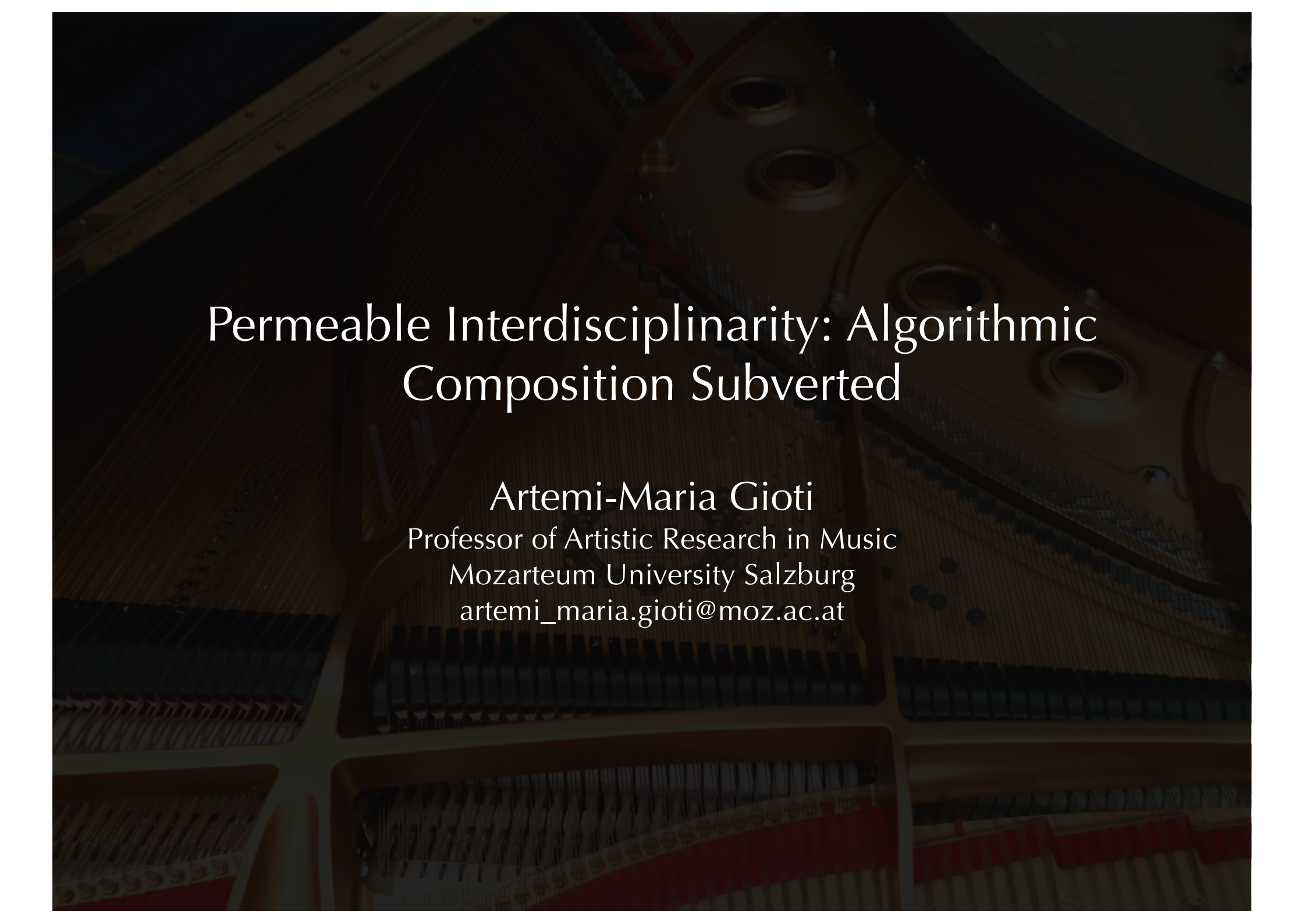




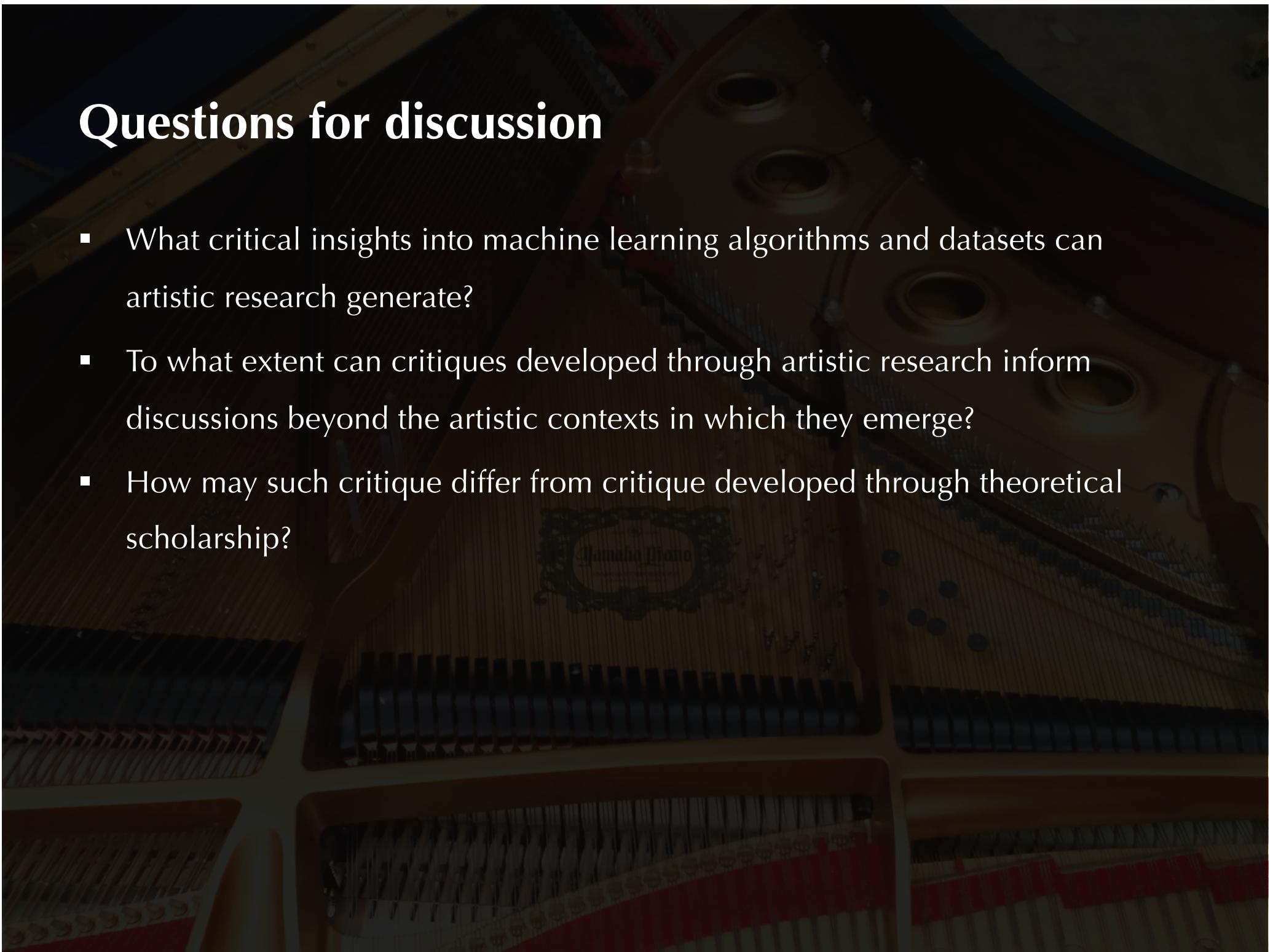
# Permeable Interdisciplinarity: Algorithmic Composition Subverted

Artemi-Maria Gioti

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Mozarteum University Salzburg  
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# Questions for discussion

- What critical insights into machine learning algorithms and datasets can artistic research generate?
- To what extent can critiques developed through artistic research inform discussions beyond the artistic contexts in which they emerge?
- How may such critique differ from critique developed through theoretical scholarship?





# BIAS II

for piano and interactive music system

Artemi-Maria Gioti  
(2022)

Cluster 1

Cluster 1 consists of four musical staves, each with a specific instruction and dynamic marking:

- Staff 1: *mf*. Instruction: "Rub along the strings." (Icon: hand rubbing strings)
- Staff 2: *mf*. Instruction: "Place one edge of the triangle in-between the strings and slide along them." (Icon: triangle on strings)
- Staff 3: *pp*. Instruction: "Place one edge of the triangle in-between the strings and slide along them." (Icon: triangle on strings)
- Staff 4: *mp*. Instruction: "Rub along the tape in a short, fast motion." (Icon: hand rubbing tape)

Cluster 2

Cluster 2 consists of three musical staves, each with a specific instruction and dynamic marking:

- Staff 1: *mf*. Instruction: "Rub along the surface." (Icon: hand rubbing surface)
- Staff 2: *mf*. Instruction: "Rub along the surface." (Icon: hand rubbing surface)
- Staff 3: *mf*. Instruction: "Rub along the silicon stick." (Icon: hand rubbing silicon stick)

Cluster 3

Cluster 3 consists of four musical staves, each with a specific instruction and dynamic marking:

- Staff 1: *mf*. Instruction: "Rub along the strings." (Icon: hand rubbing strings)
- Staff 2: *mf*. Instruction: "Place one edge of the triangle in-between the strings and slide along them." (Icon: triangle on strings)
- Staff 3: *mf*. Instruction: "Place one edge of the triangle in-between the strings and slide along them." (Icon: triangle on strings)
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Cluster 4

Cluster 4 consists of two musical staves, each with a specific instruction and dynamic marking:

- Staff 1: *mf*. Instruction: "Rub along the surface." (Icon: hand rubbing surface)
- Staff 2: *mf*. Instruction: "Rub along the surface." (Icon: hand rubbing surface)

Cluster 5

Cluster 5 consists of two musical staves, each with a specific instruction and dynamic marking:

- Staff 1: *mf*. Instruction: "Place the magnet on top of the string with its strong side facing away from it and let it vibrate." (Icon: magnet on string)
- Staff 2: *mf*. Instruction: "Place the magnet on top of the string with its strong side facing away from it and let it vibrate." (Icon: magnet on string)

Cluster 6

Cluster 6 consists of two musical staves, each with a specific instruction and dynamic marking:

- Staff 1: *mf*. Instruction: "Press and release one of the clips. Let them oscillate." (Icon: hand pressing clip)
- Staff 2: *mf*. Instruction: "Rub along the strings." (Icon: hand rubbing strings)

Cluster 7

Cluster 7 consists of three musical staves, each with a specific instruction and dynamic marking:

- Staff 1: *mf*. Instruction: "Place one edge of the triangle in-between the strings and slide along them." (Icon: triangle on strings)
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# BIAS II

for piano and interactive music system

Artemi-Maria Gioti  
(2022)

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- Staff 1: Rub along the strings. *mf*
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- Staff 1: *mf*
- Staff 2: *mf*
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Cluster 4 contains two musical staves with specific instructions:

- Staff 1: *mf*
- Staff 2: *mf*

Cluster 5

Cluster 5 contains two musical staves with specific instructions:

- Staff 1: Place the magnet on top of the string with its strong side facing away from it and let it vibrate.
- Staff 2: *mf*

Cluster 6

Cluster 6 contains two musical staves with specific instructions:

- Staff 1: Press and release one of the clips. Let them oscillate.
- Staff 2: Rub along the strings.

Cluster 7

Cluster 7 contains three musical staves with specific instructions:

- Staff 1: *mf*
- Staff 2: *mf*
- Staff 3: *mf*



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for piano and interactive music system

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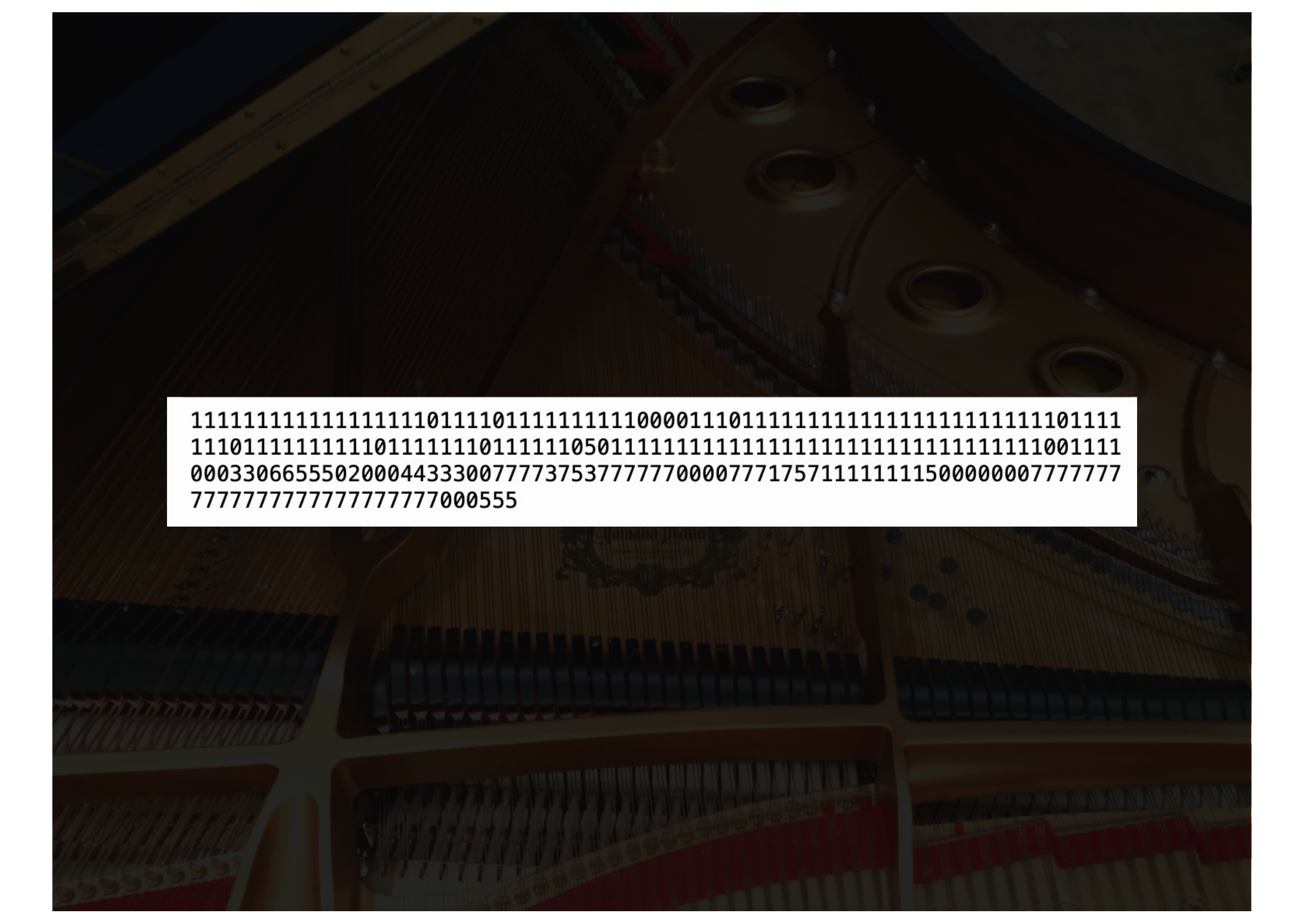
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## Main themes in *Bias II*

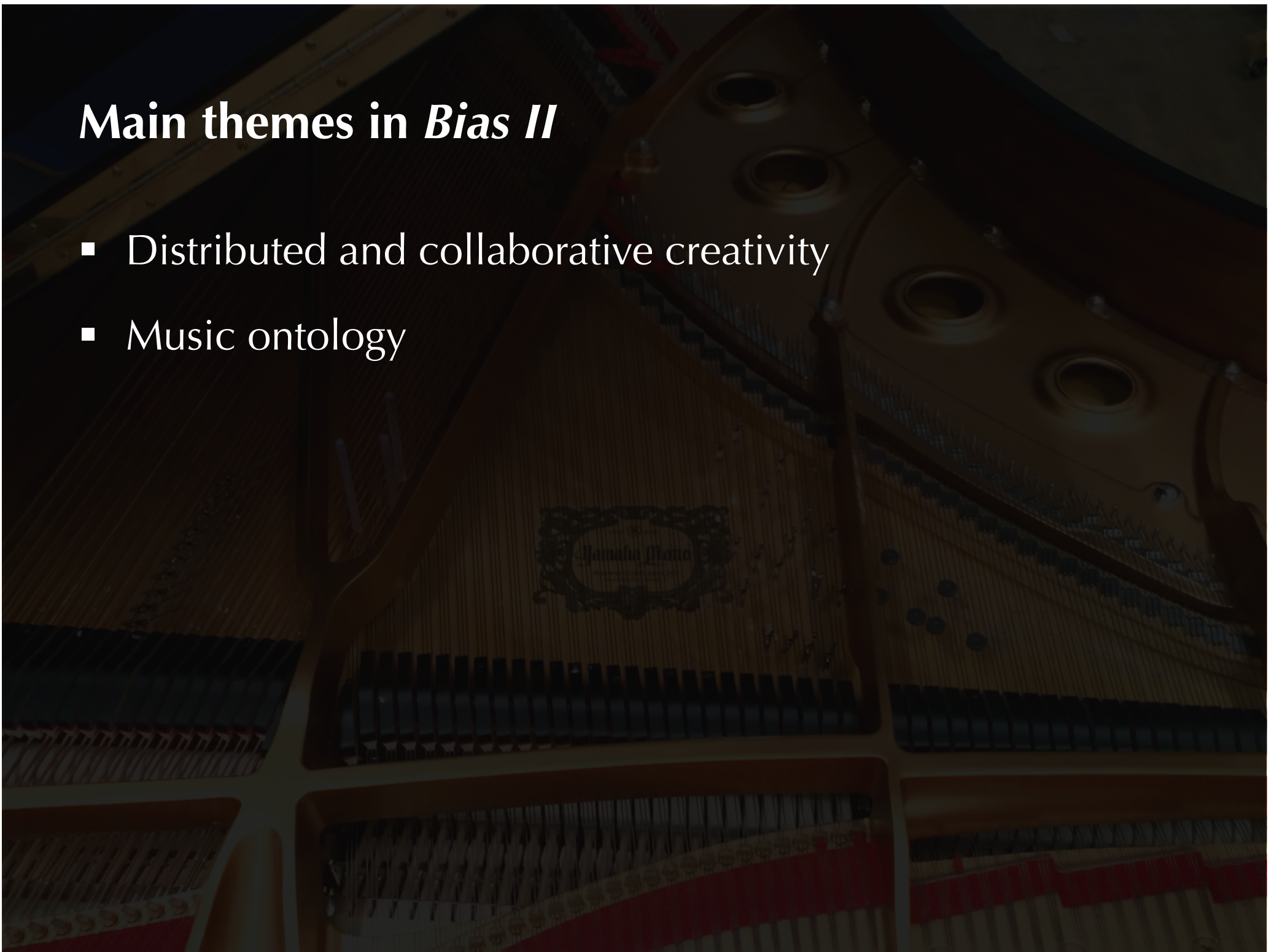
- Distributed and collaborative creativity
- Music ontology
  - "provisional musical work which both retains and blurs the traces and boundaries of individual and collective authorship" (Born 2005)





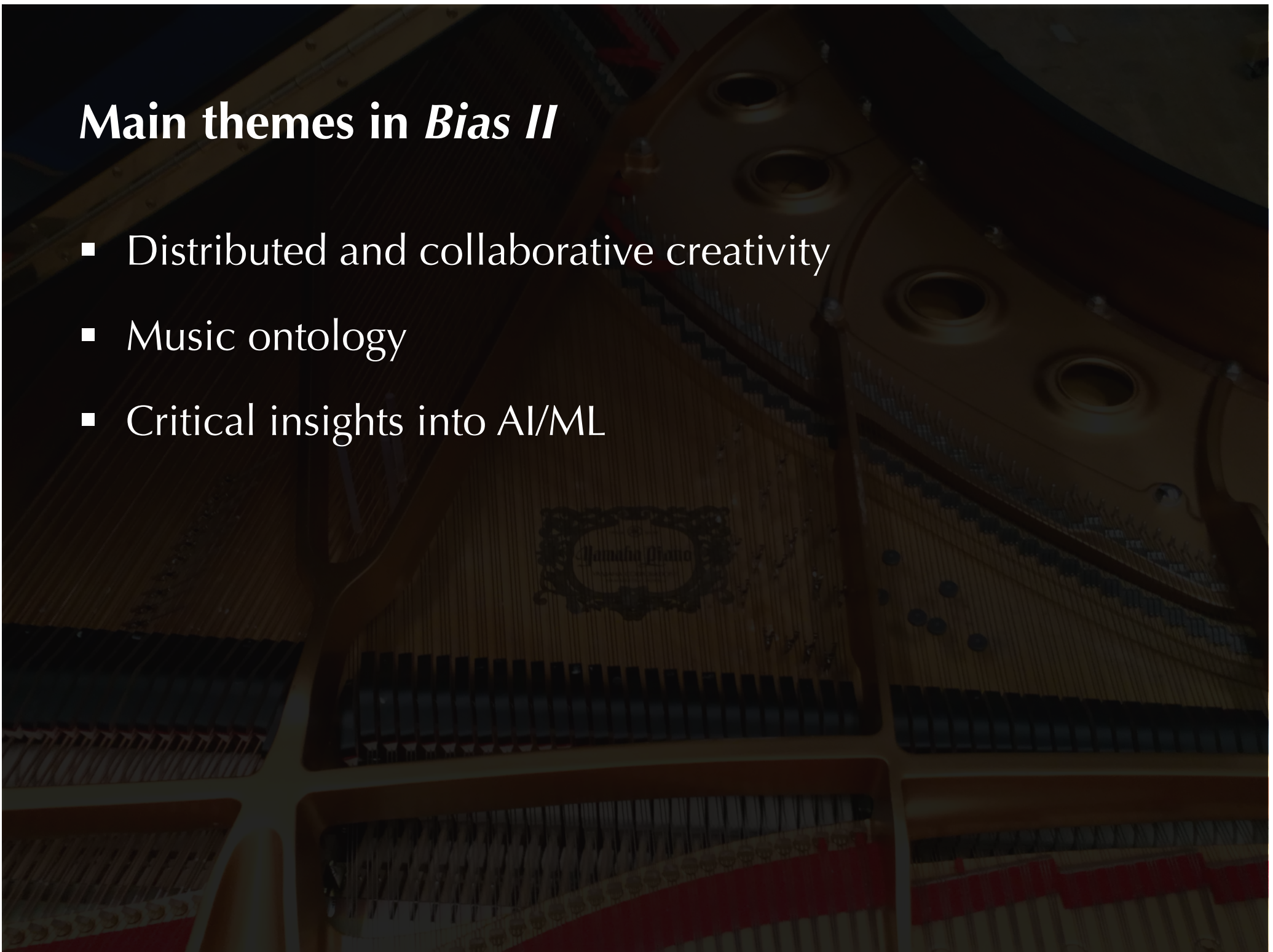
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- Distributed and collaborative creativity
- Music ontology



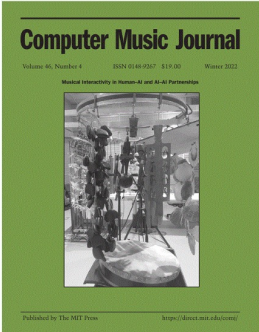
## Main themes in *Bias II*

- Distributed and collaborative creativity
- Music ontology
- Critical insights into AI/ML





Volume 46, Issue 4  
Winter 2022



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- Case Study 2: Bias II for Piano and Interactive Music System
- Conclusions
- Acknowledgments
- References

December 01 2022

**Composing the Assemblage: Probing Aesthetic and Technical Dimensions of Artistic Creation with Machine Learning** 🔑

Artemi-Maria Gioti, Aaron Einbond, Georgina Born

> Author and Article Information

*Computer Music Journal* (2022) 46 (4): 62–80.

[https://doi.org/10.1162/comj\\_a\\_00658](https://doi.org/10.1162/comj_a_00658)

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Abstract

In this article, we address the role of machine learning (ML) in the composition of two new musical works for acoustic instruments and electronics through autoethnographic reflection on the experience. Our study poses the key question of how ML shapes, and is in turn shaped by, the aesthetic commitments characterizing distinctive compositional practices. Further, we ask how artistic research in these practices can be informed by critical themes from humanities scholarship on material engagement and critical data studies. Through these frameworks, we consider in what ways the interaction with ML algorithms as part of the compositional process differs from that with other music technology tools. Rather than focus on narrowly conceived ML algorithms, we take into account the heterogeneous assemblage brought into play: from composers, performers, and listeners to loudspeakers, microphones, and audio descriptors. Our analysis focuses on a deconstructive critique of data as being contingent on the decisions and material conditions involved in the data-creation process. It also explores how interaction among the human and nonhuman collaborators in the ML assemblage has significant similarities to – as well as differences from – existing models of material engagement. Tracking the creative process of composing these works, we uncover the aesthetic implications of the many nonlinear collaborative decisions involved in composing the assemblage.

Research in music and AI is, by definition, an interdisciplinary endeavor. Yet, the field of music AI seems to be leaning heavily towards the technoscientific side of its disciplinary genealogy, with

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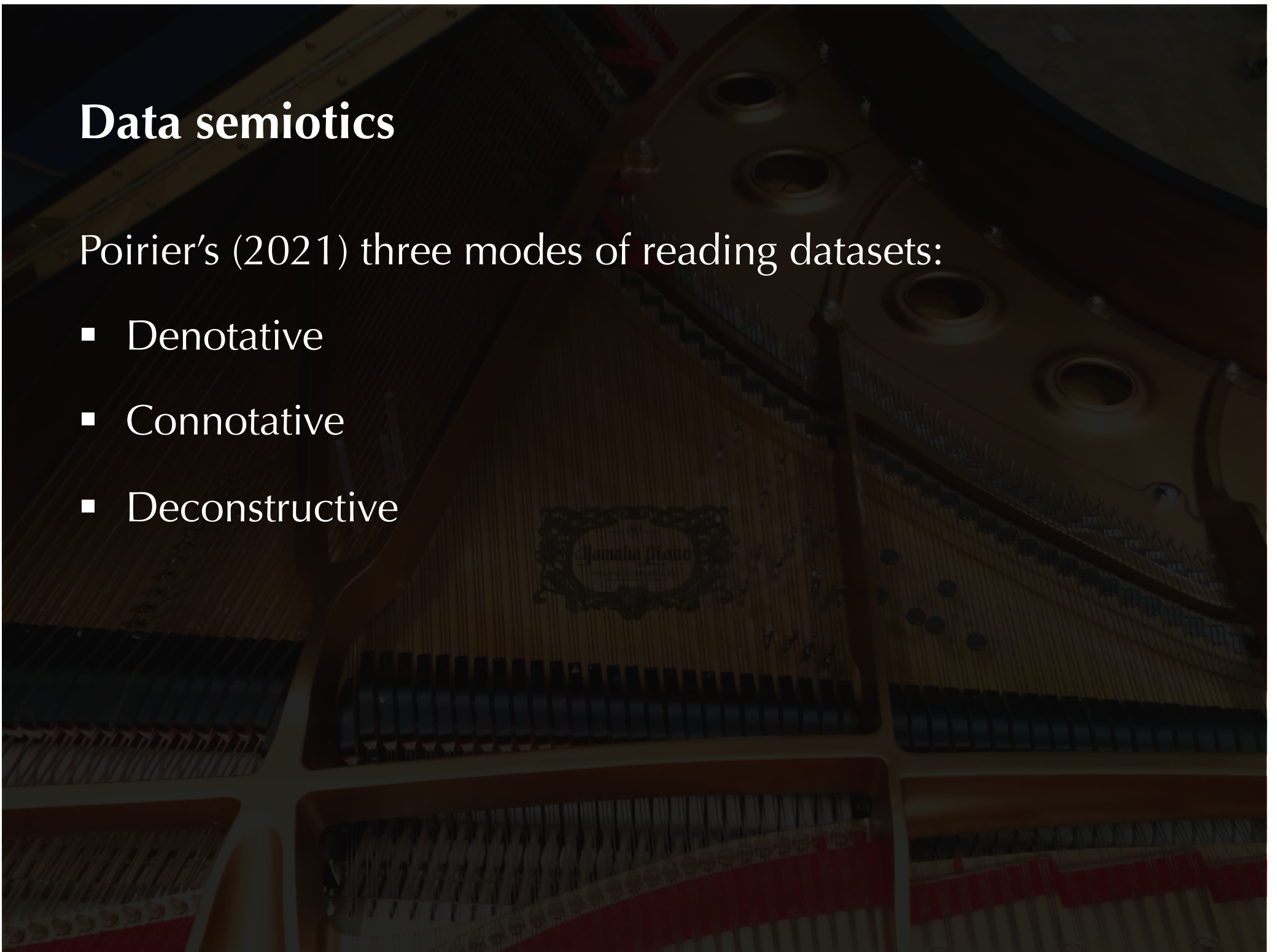
Web of Science



# Data semiotics

Poirier's (2021) three modes of reading datasets:

- Denotative
- Connotative
- Deconstructive

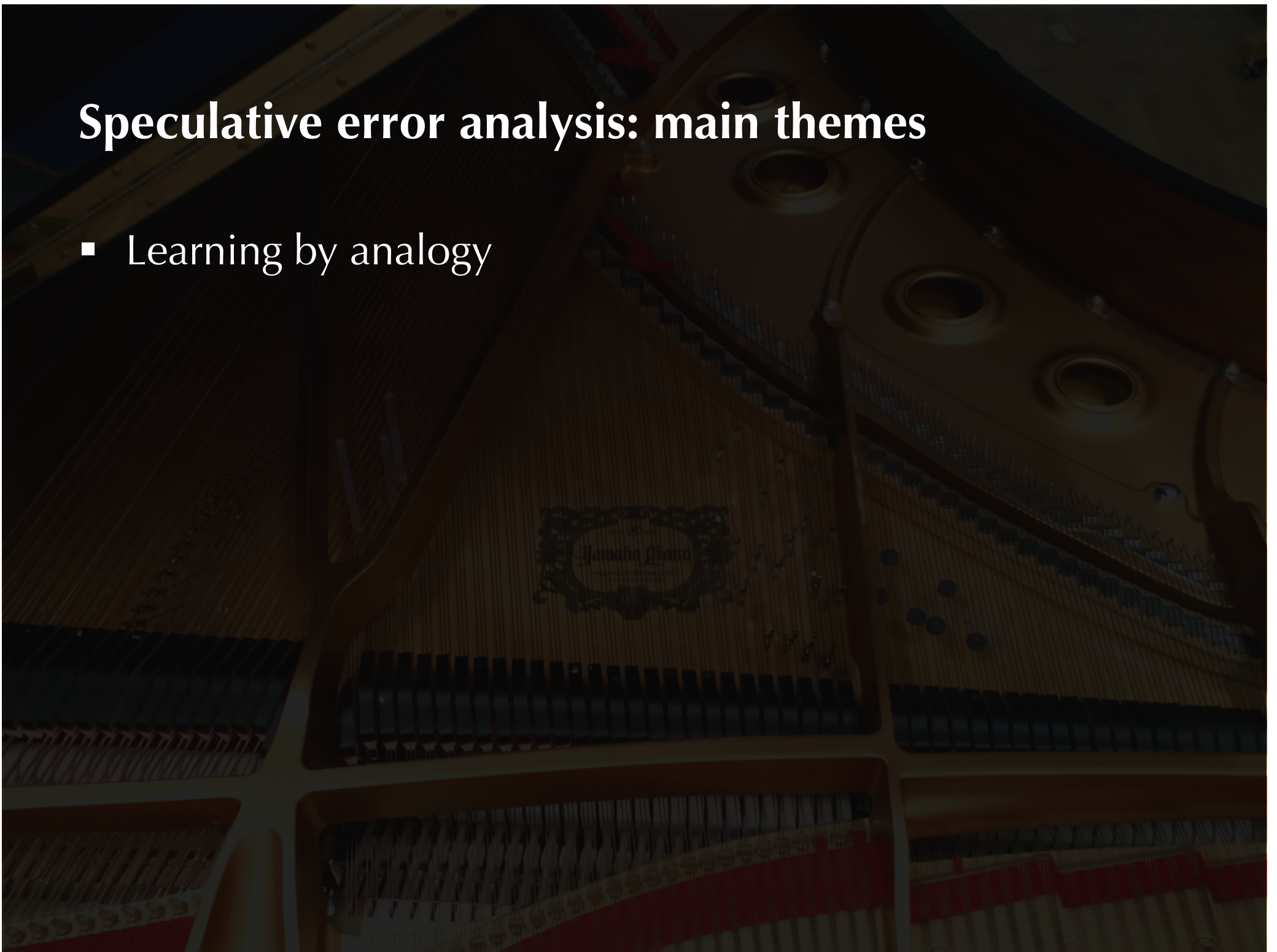




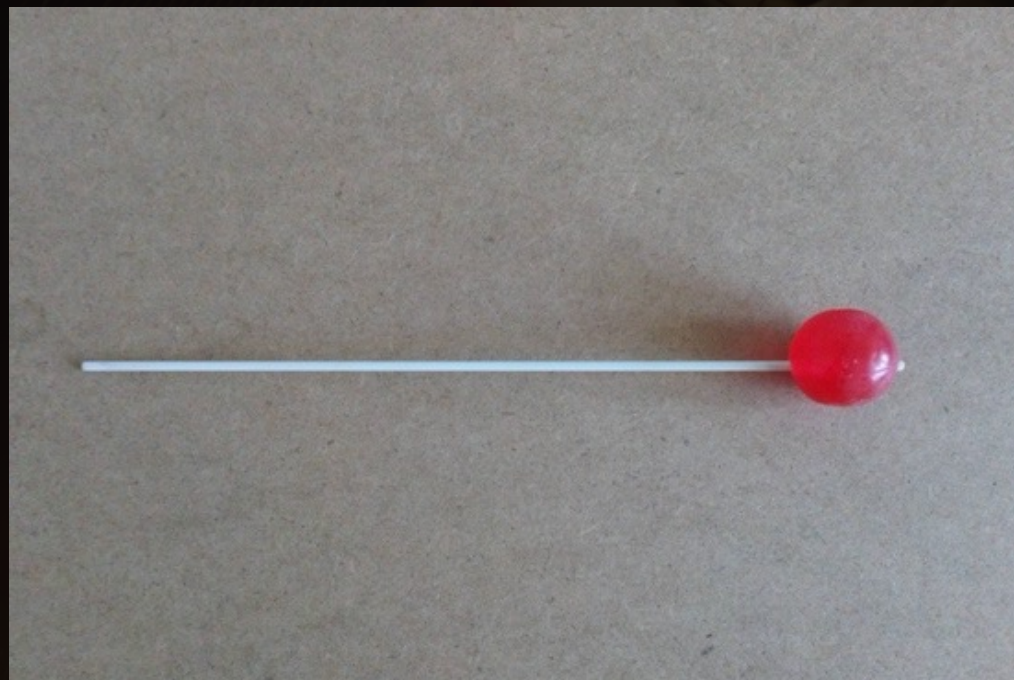
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# Speculative error analysis: main themes

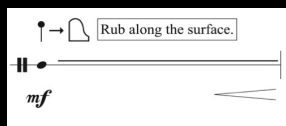
- Learning by analogy







*"Superball" mallet*



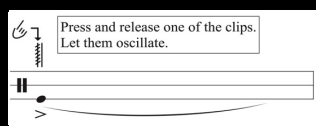
*Cluster 2: training set*



*Cluster 2: David's interpretation*



*Cluster 5: training set*



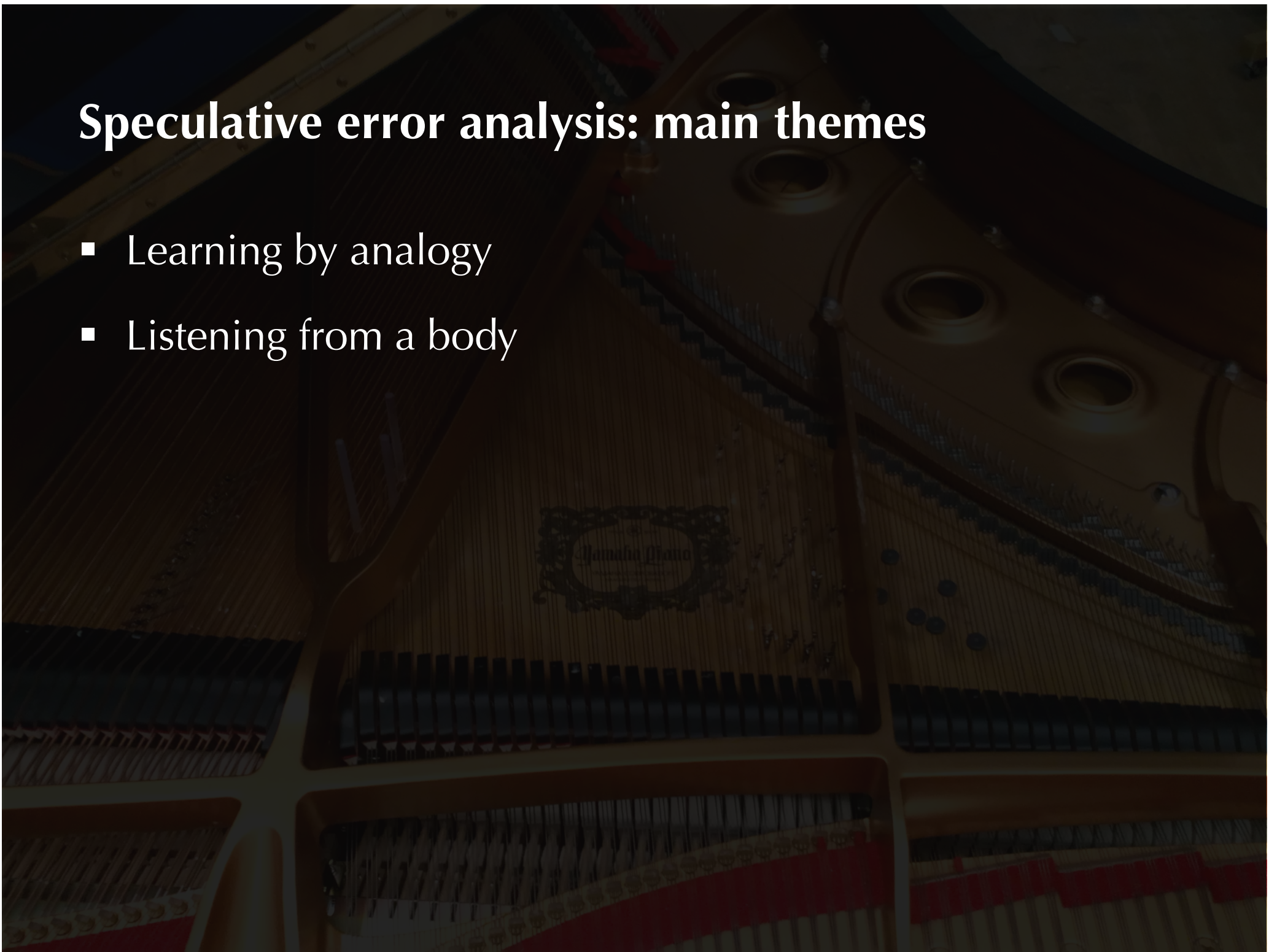
*Cluster 6: training set*





# Speculative error analysis: main themes

- Learning by analogy
- Listening from a body



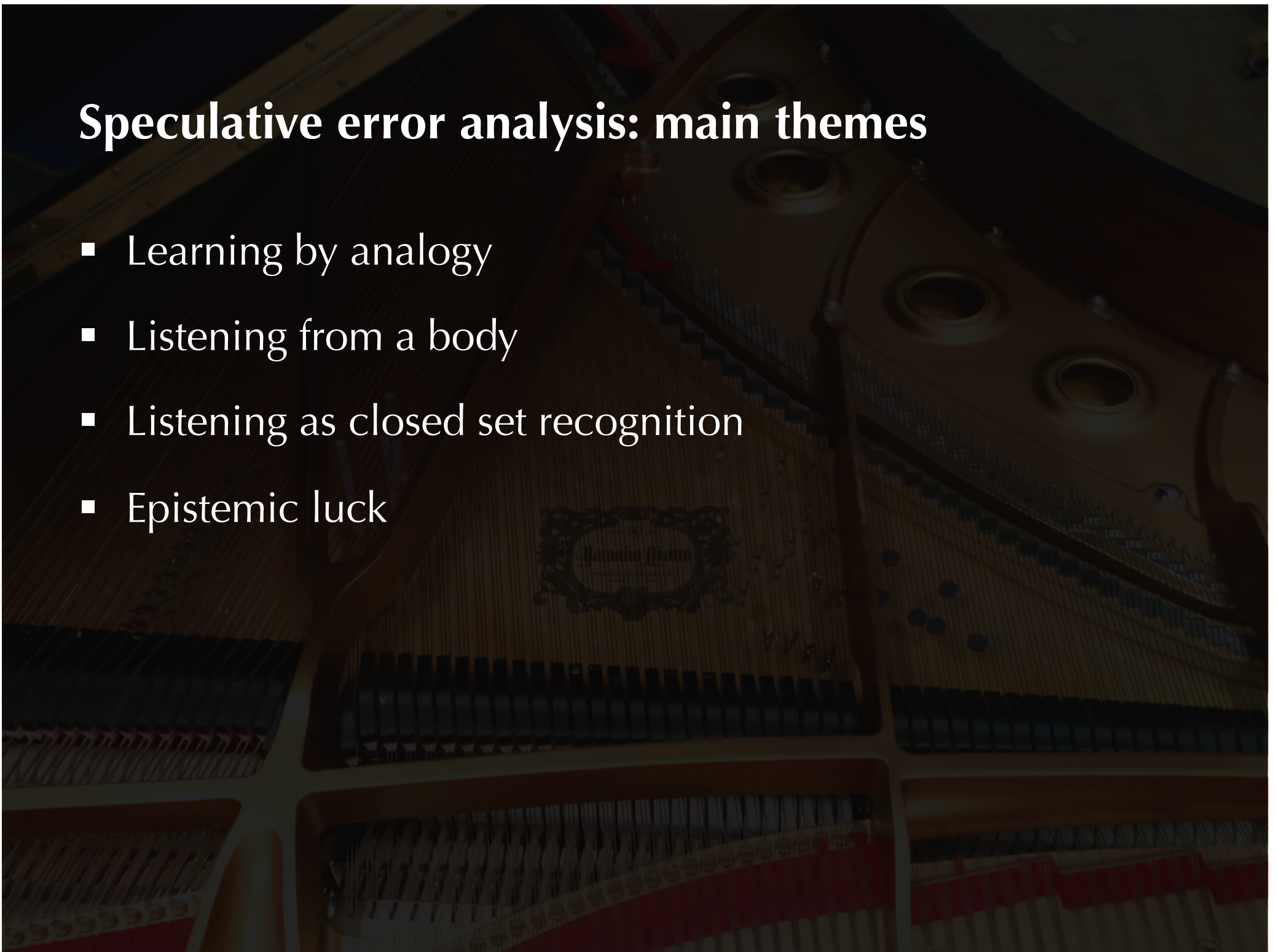






# Speculative error analysis: main themes

- Learning by analogy
- Listening from a body
- Listening as closed set recognition
- Epistemic luck

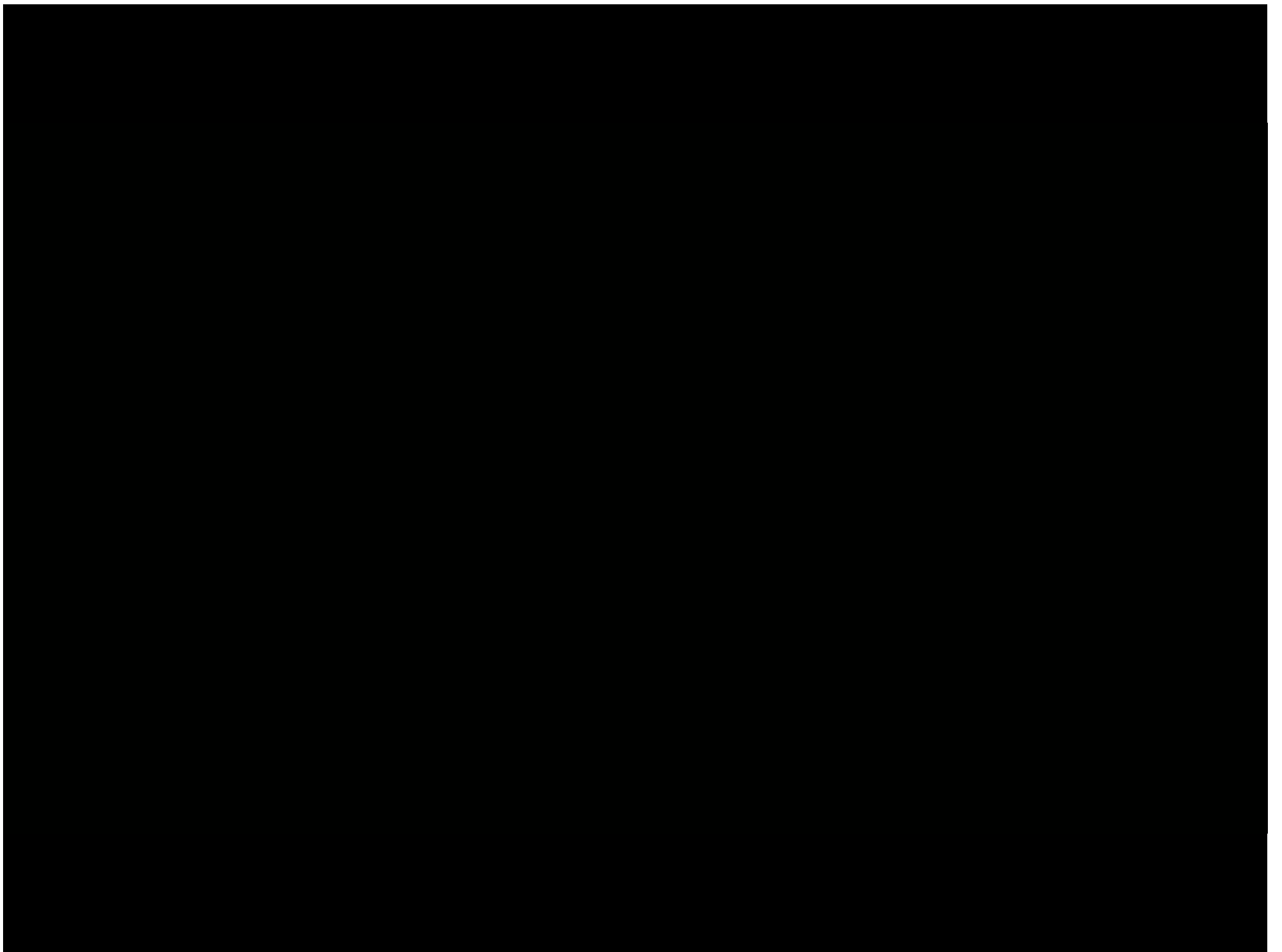




Crocodile-clip prepared strings









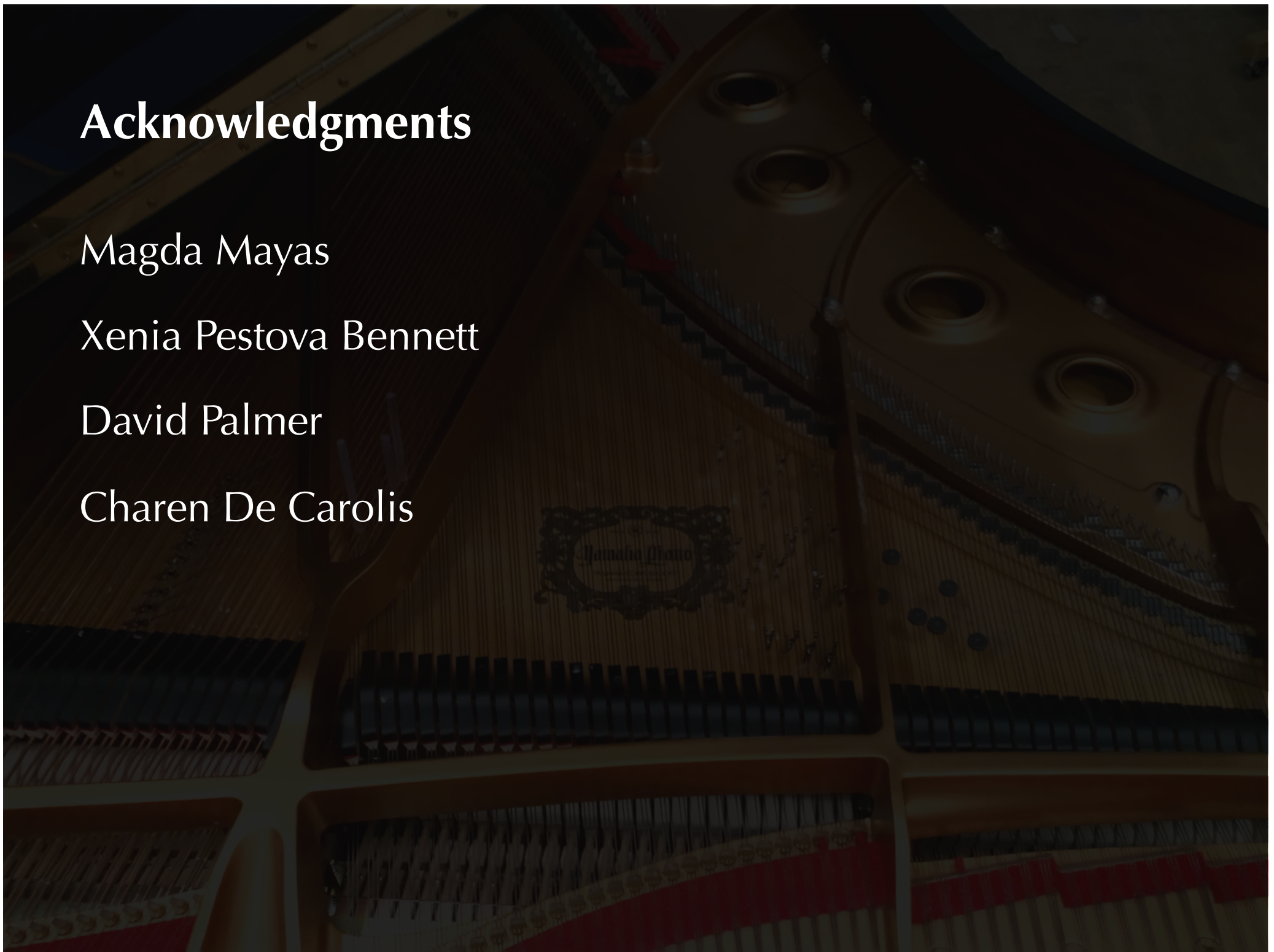
# Acknowledgments

Magda Mayas

Xenia Pestova Bennett

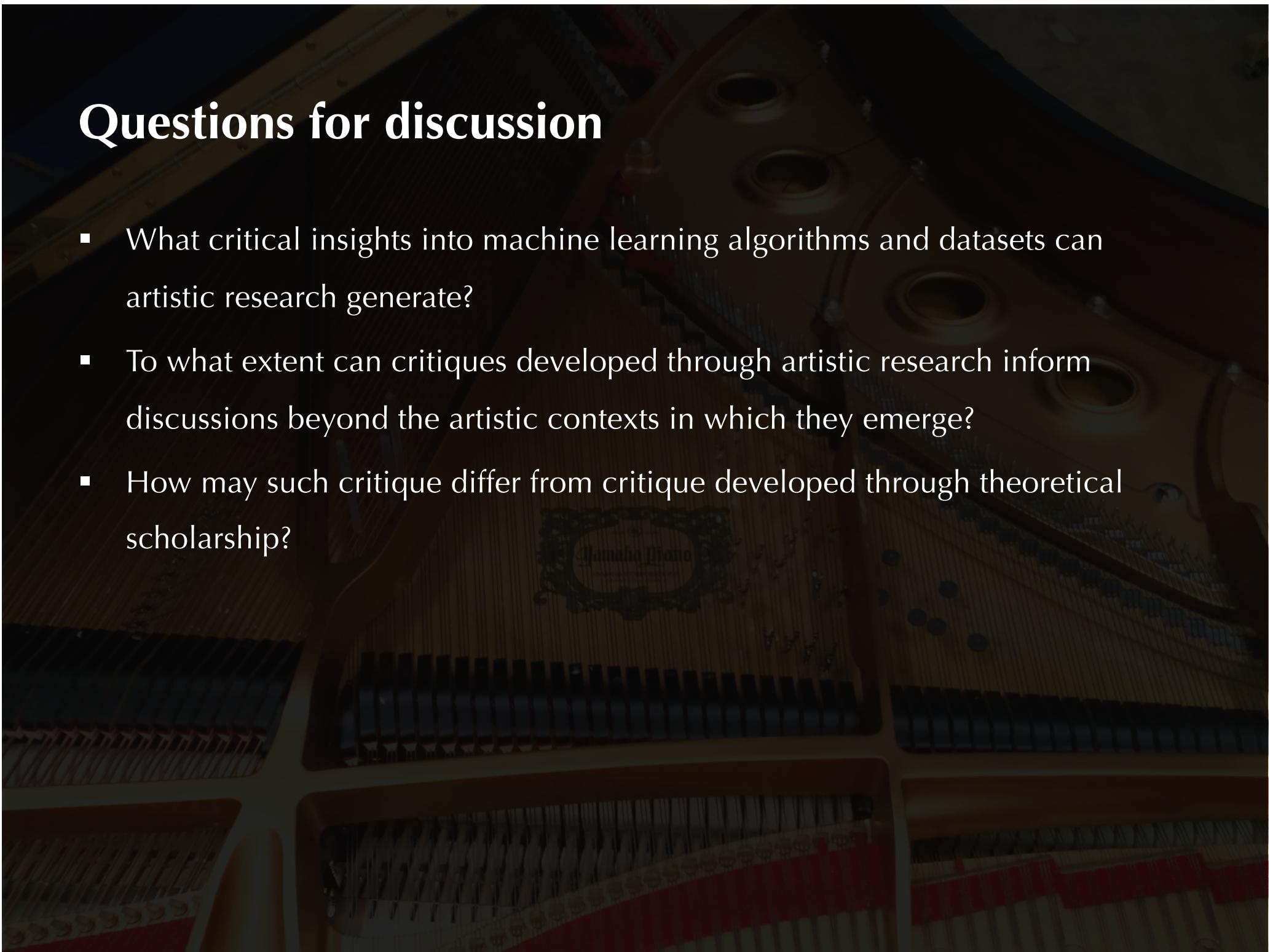
David Palmer

Charen De Carolis



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